

Machine Learning for Biology and Health

CSCI 1851
Spring 2026

Ritambhara Singh

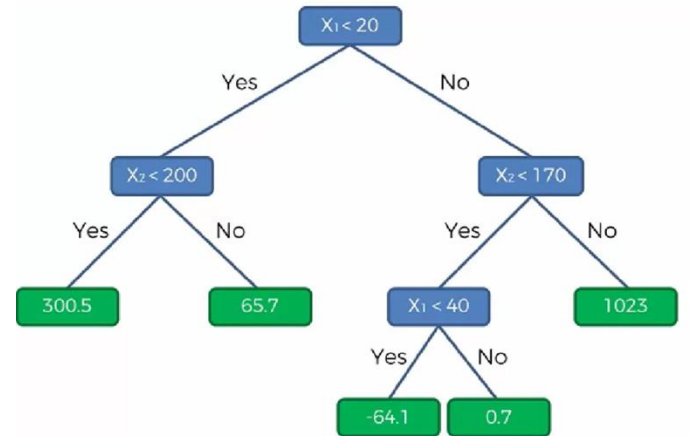
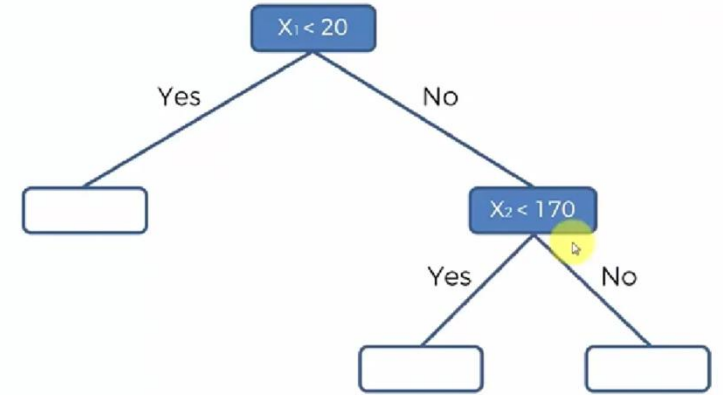
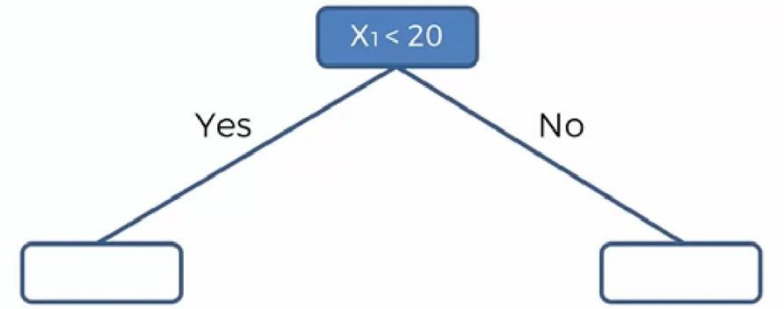
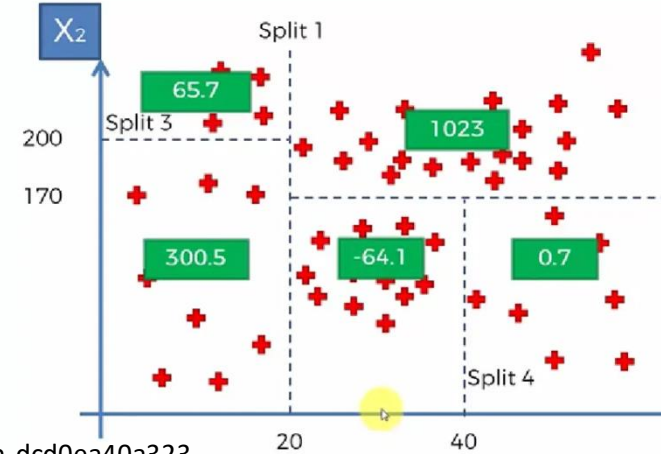
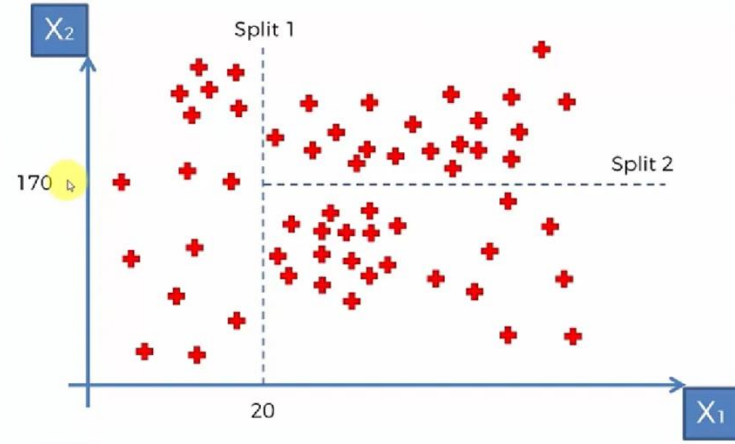
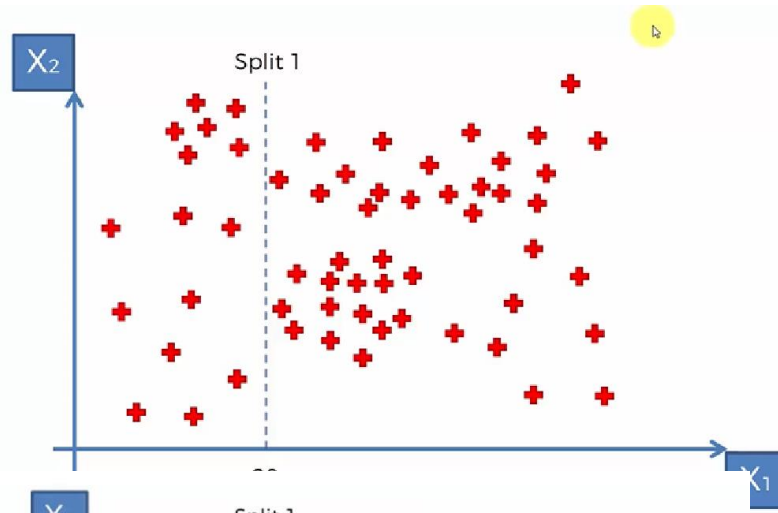
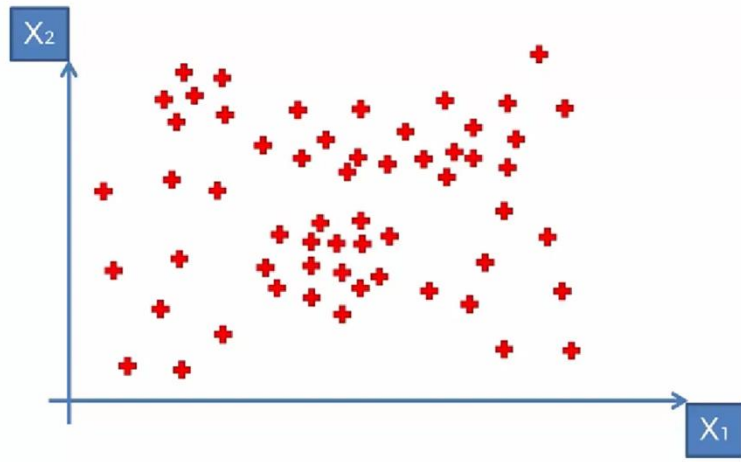
February 12, 2026
Thursday



Classification and Regression Tree (CART) Algorithm

Start with a top-level parent node that contains all the training data observations. Then:

1. Conduct a **greedy search** for the “best” split of the parent node by
 1. Identifying a range of reasonable possible splits each consisting of a binary question involving a feature and threshold.
 2. For each potential split option identified in the previous step, compute the variance of the responses (for continuous responses) or Gini impurity (for binary responses) in each of the resulting child nodes.
 3. Identify which of the potential splits leads to the lowest variance measure (for continuous responses) or Gini impurity measure (for binary responses) of the child nodes. This is the “best” split.
 2. Implement the best split identified in the previous step to create two child nodes.
 3. For each resulting child node that does not meet a prespecified stopping criteria, repeat steps 1 and 2 (i.e., each child node is now the parent node and will be split into subsequent child nodes).
- If a child node meets the prespecified stopping criteria do not split the child node. A child node that is not split further is called a “**leaf node**” or “**terminal node**”.
4. Continue computing sequential splits until all child nodes meet the prespecified stopping criteria.



Random Forest Learning Algorithm



Randomness is introduced into each tree by

- Training each tree using a different random bootstrap sample of the training data.
- Considering a different random subset of the predictor variables for each node split.

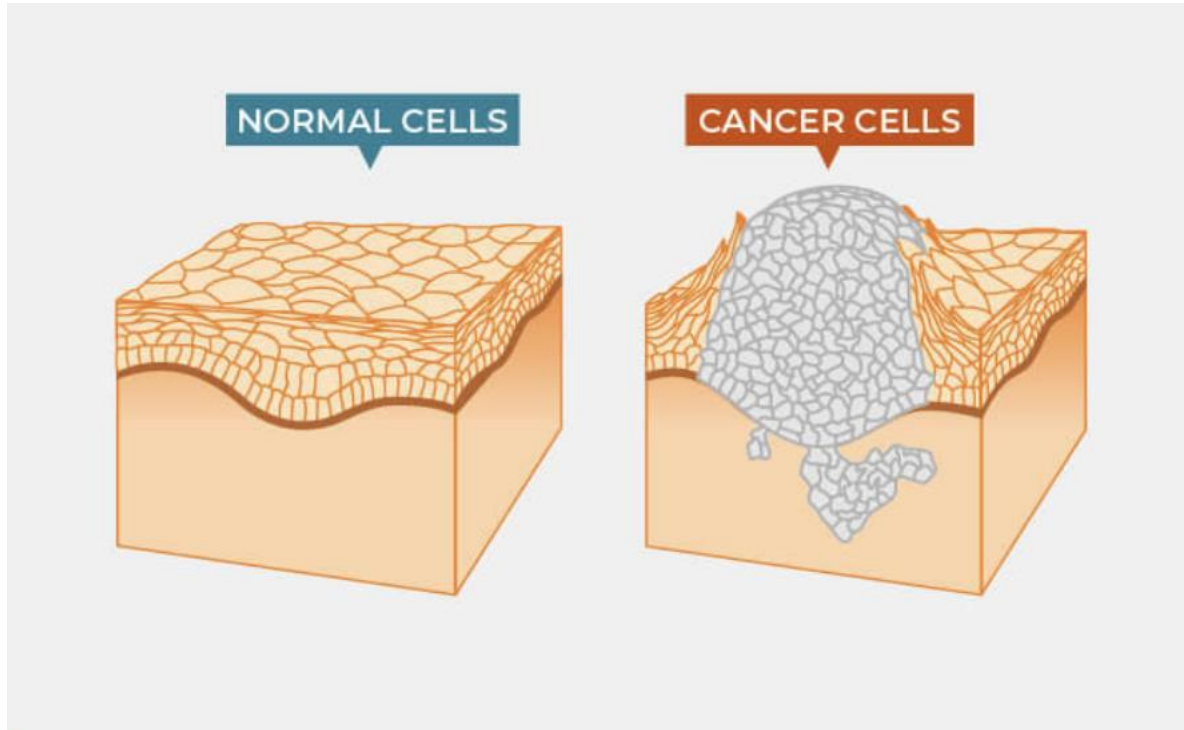
This option can be implemented for the decision tree as well

Today's goal - Learn about multi-class classification and boosting

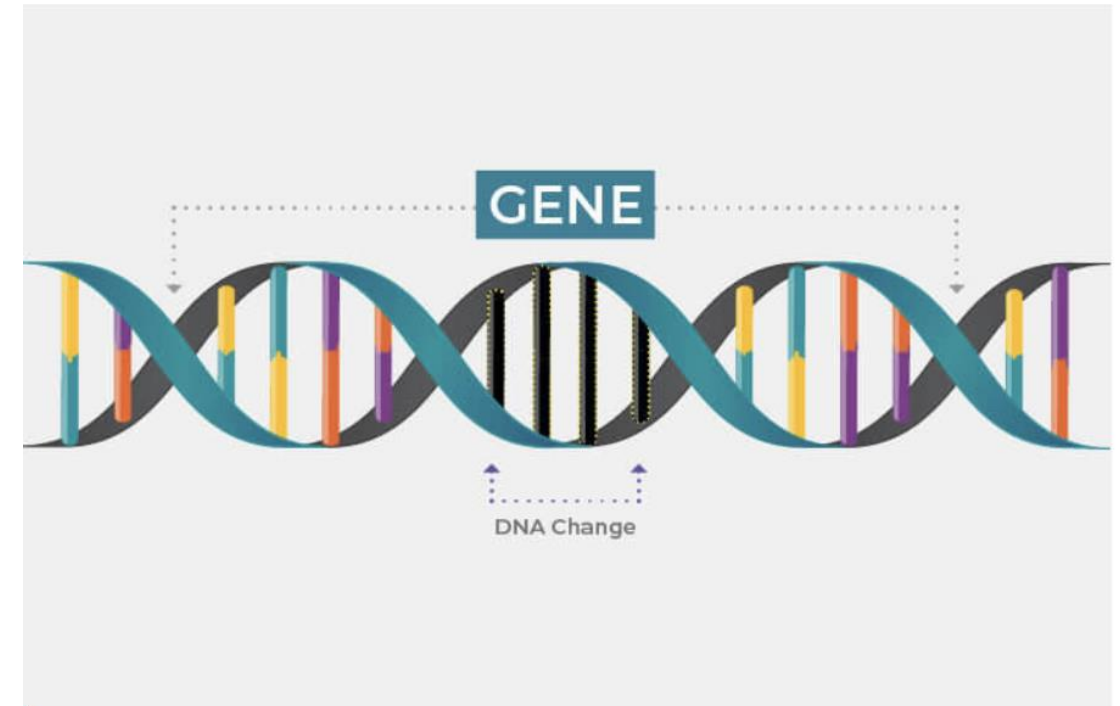
- (1) Introducing the task – Predicting cancer-type using gene expression
- (2) Multi-class classification
- (3) Class Activity: Multi-class classification evaluation
- (4) Boosting
- (5) Gradient Boosted Trees

Predicting cancer-type using gene expression

Studying Cancer



Cancer is a disease caused when cells divide uncontrollably and spread into surrounding tissues.

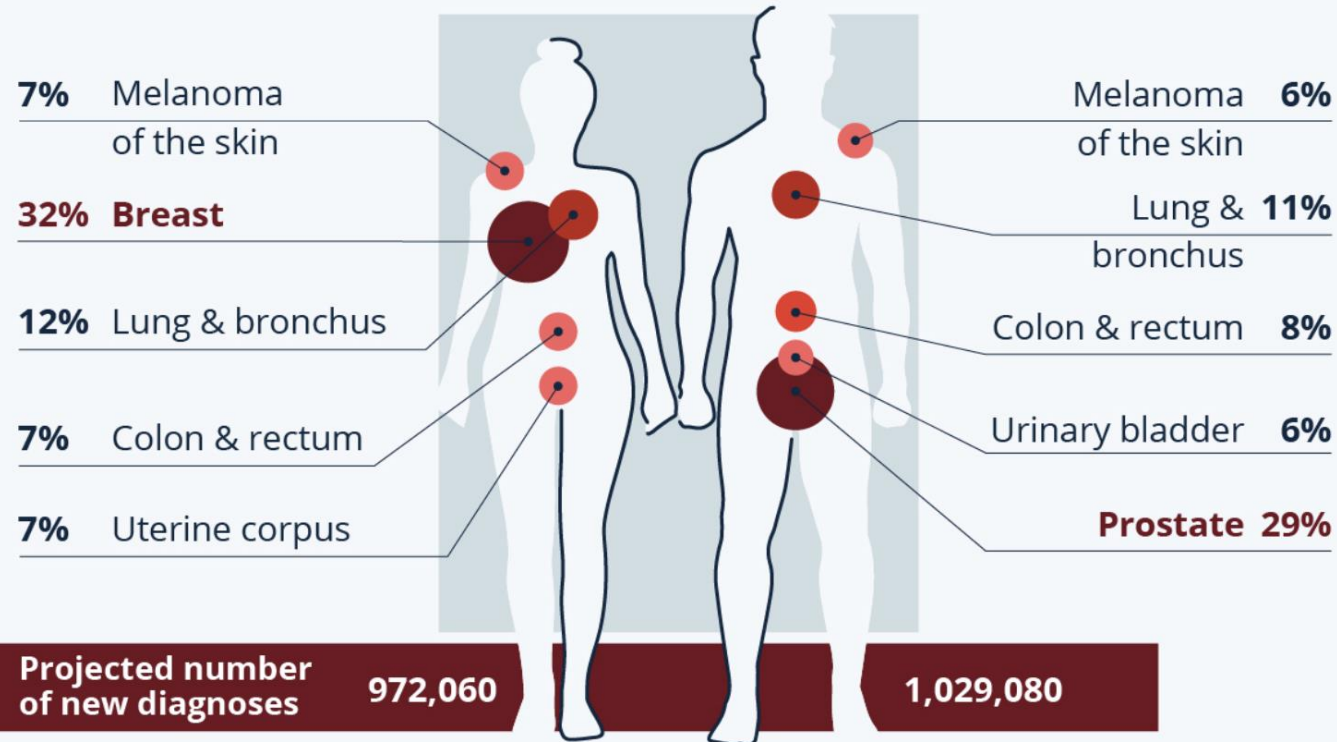


Cancer is caused by changes to DNA. Most cancer-causing DNA changes occur in sections of DNA called genes. These changes are also called genetic changes.

Studying Cancer

The Most Common Types of Cancer in the U.S.

Projected share of new cancer diagnoses in the U.S. in 2024, by gender



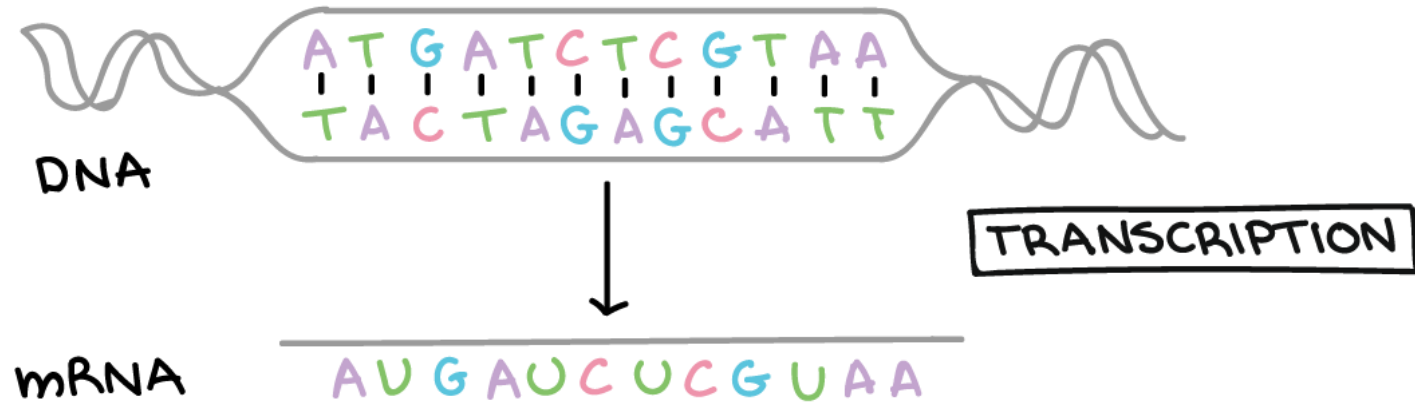
Source: American Cancer Society

Disclaimer: I am
NOT a biologist!

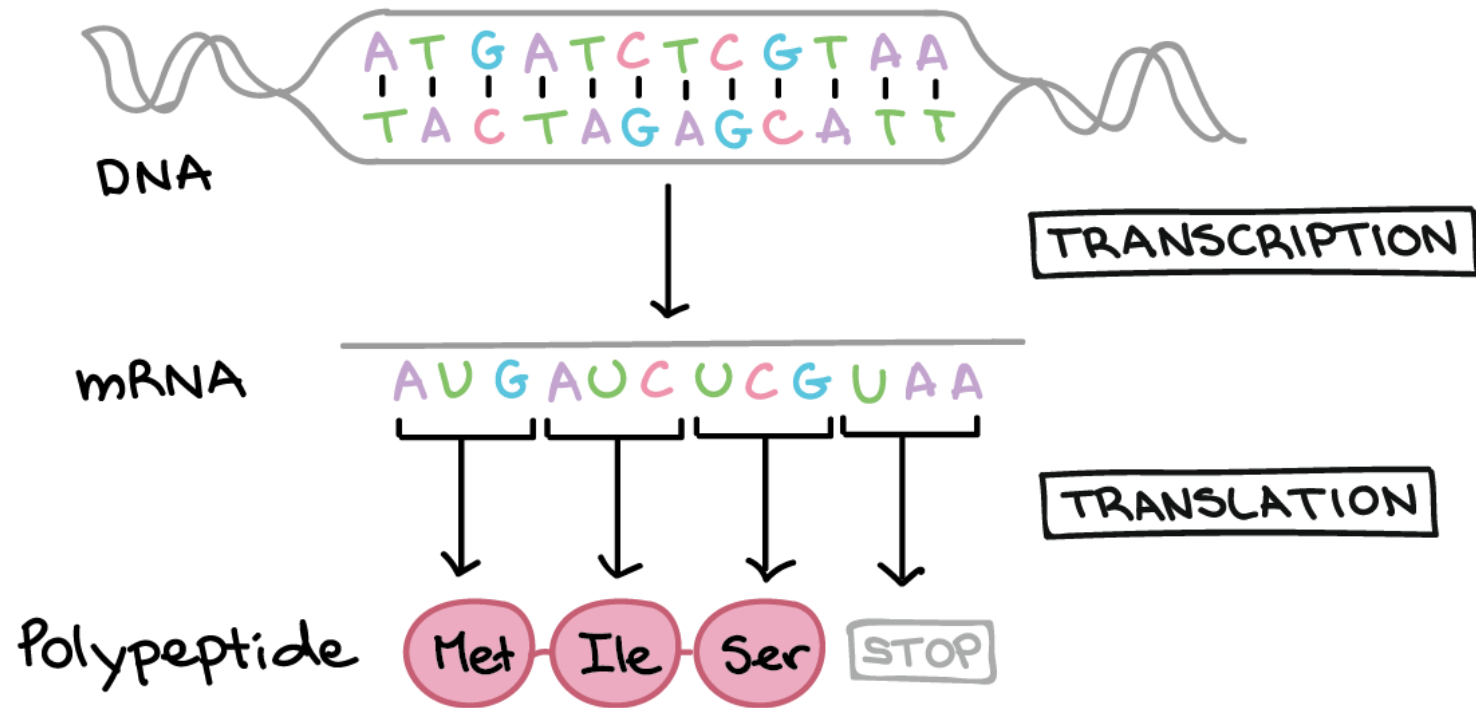
THE CENTRAL DOGMA



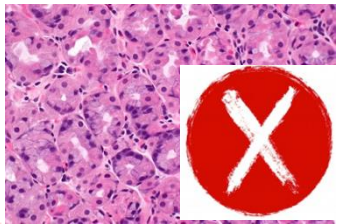
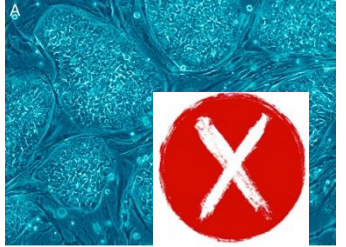
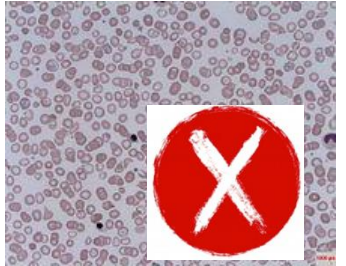
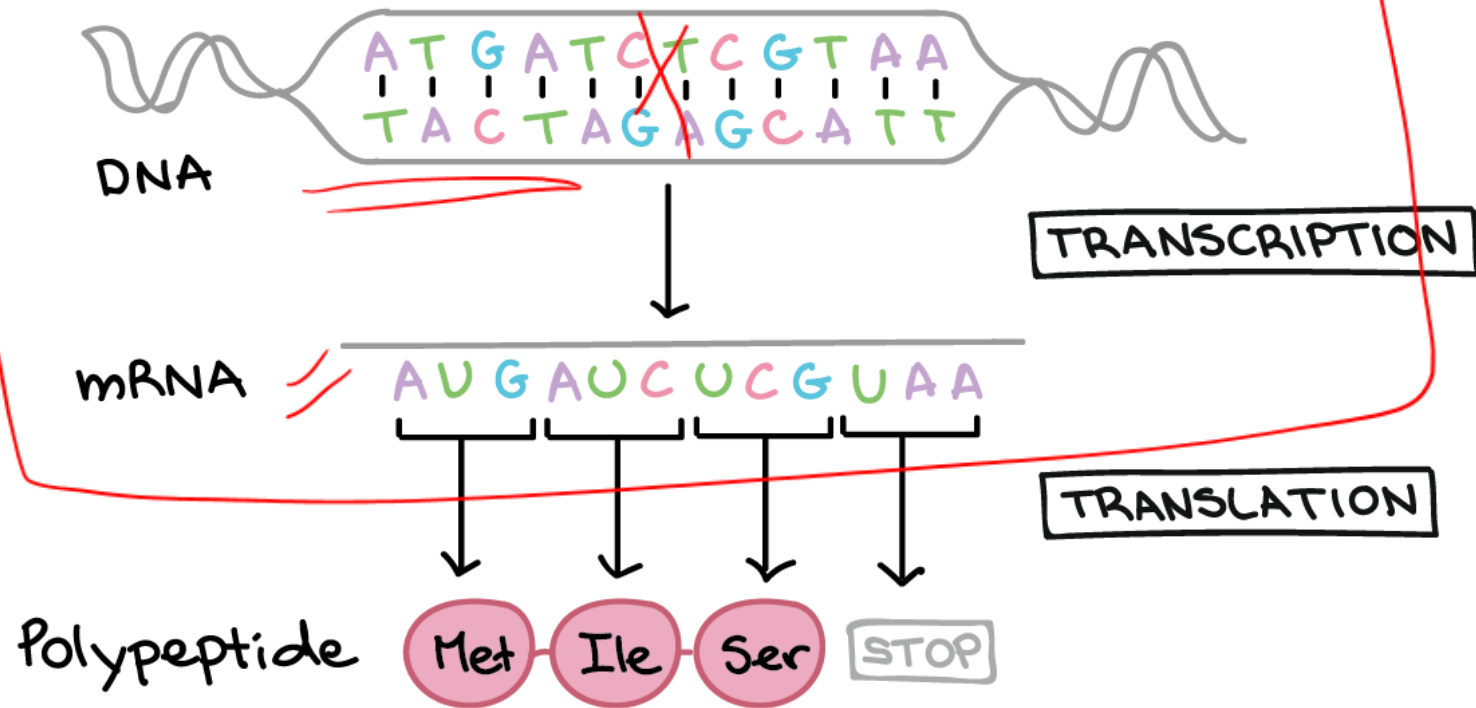
THE CENTRAL DOGMA



THE CENTRAL DOGMA



THE CENTRAL DOGMA



6

How do we view the DNA

~ 3 Billion base pairs (bp)

GGCGAGTCCGCGGGGGAGGCGGGCAGAGCCTGAGCTCAGGTCTTTCTGCGTCTGGCGGA GGCGAGTCCGCGGGGGAGGCGGG

Reference genome

Main strand	A	T	G	A	T	T	G	A	C	A	T	T	G	A	G	G	A	T	C	C	A	T
Complementary Strand	T	A	C	T	A	A	C	T	G	T	A	A	C	T	C	C	T	A	G	G	T	A

Sample genetic code with complementary strands.

© G.Osuri

How do we view the DNA

What is a gene?

Coordinates: chr1: 14569-14780

Can we get any other information about the gene?

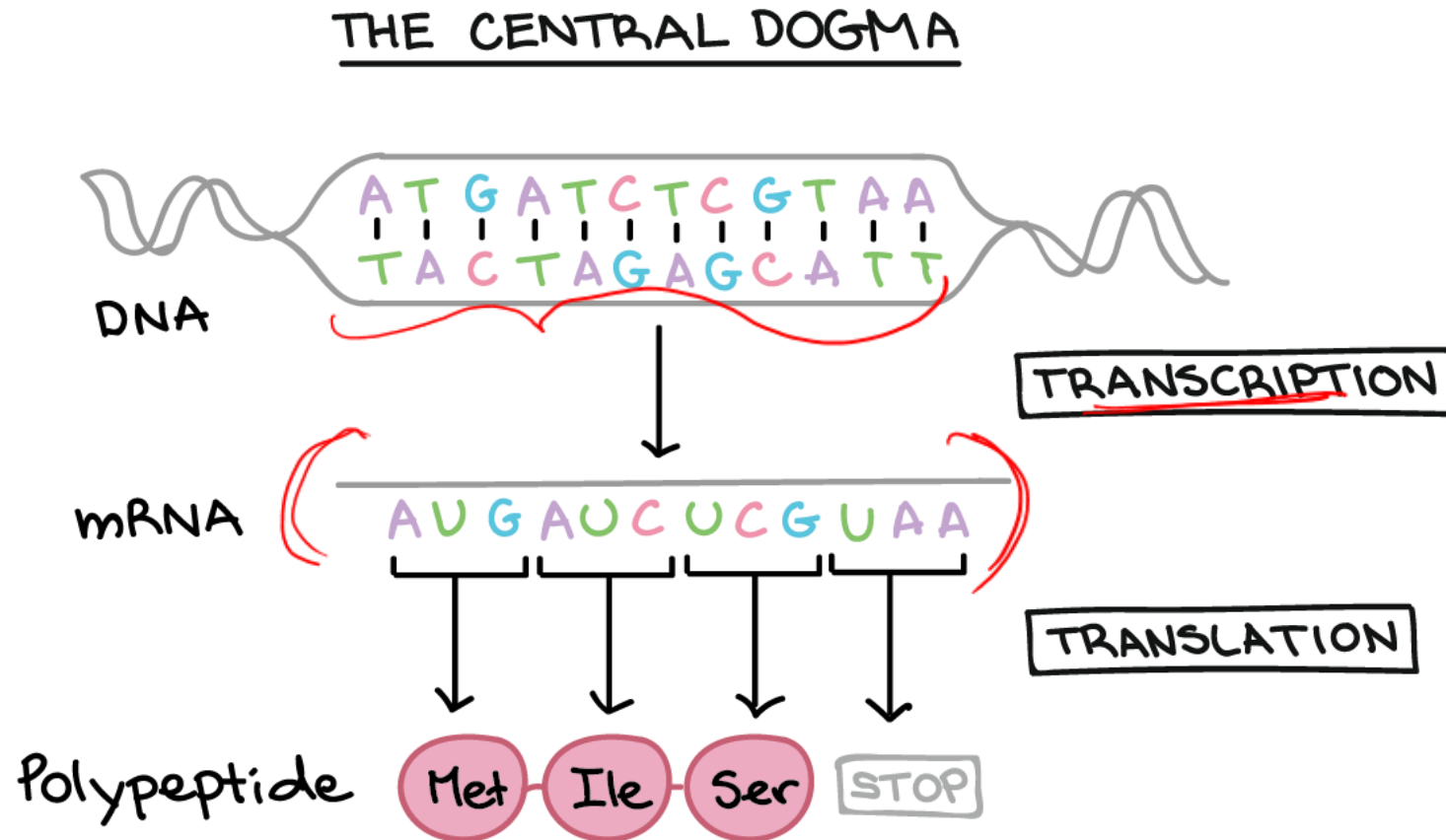
Gene

FASTA file

→ NC_008679.1:5001-38170 Homo sapiens paired box 6 (PAX6)

ACCTCTTTTCTTATCAATTGACATTTAAACTCTGGGGCAGGTCTCGCGTAGAACGCGGCTGTCAGATCT
GCCACTTCCCCTGCCGAGCGGCGGTGAGAAGTGTGGGAACCGGCGCTGCCAGGCTCACCTGCCTCCCCGC
CCTCCGCTCCCAGGTAACCGCCCGGGCTCCGGCCCCGGCCCGGCTCGGGGCCCGGGGCTCTCCGCTG
CCAGCGACTGCTGTCCCCAAATCAAAGCCCCGCCCAAGTGGCCCCGGGGCTTGATTTTTGCTTTTAAAAG
GAGGCATACAAAGATGGAAGCGAGTTACTGAGGGAGGGATAGGAAGGGGGGTGGAGGAGGGACTTGTCTT
TGCCGAGTGTGCTCTTCTGCAAAGTAGCAAAATGTTCCACTCCTAAGAGTGGACTTCCAGTCCGGCCCT
GAGCTGGGAGTAGGGGGCGGGAGTCTGCTGCTGCTGTCTGCTAAAGCCACTCGCGACCGCGAAAAATGCA
GGAGGTGGGGACGCACTTTGCATCCAGACCTCCTGTCATCGCAGTTCACGACATCCACGCTTGGGAAAG
TCCGTACCCGCGCCTGGAGCGCTTAAAGACACCCTGCCGCGGGTCTGGGCGAGGTGCAGCAGAAGTTTCCC
GCGGTTGCAAAGTGCAGATGGCTGGACCGCAACAAAGTCTAGAGATGGGGTTCGTTTCTCAGAAAGACGC

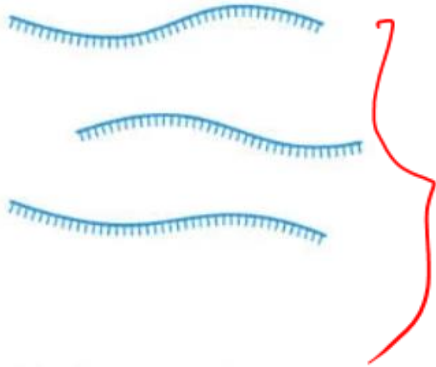
Gene expression



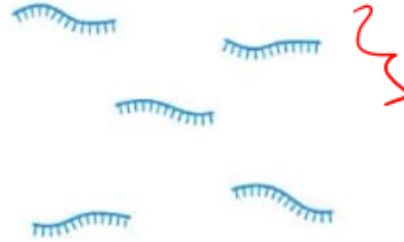
How would you
measure gene
expression?

RNA Sequencing

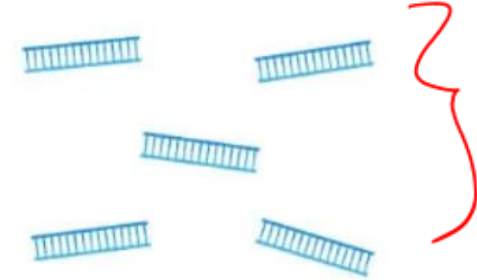
① Isolate RNA from samples



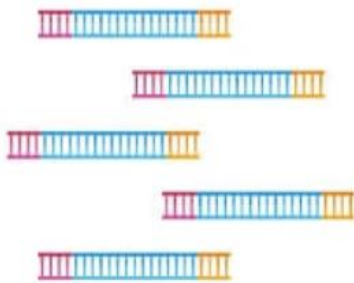
② Fragment RNA into short segments



③ Convert RNA fragments into cDNA



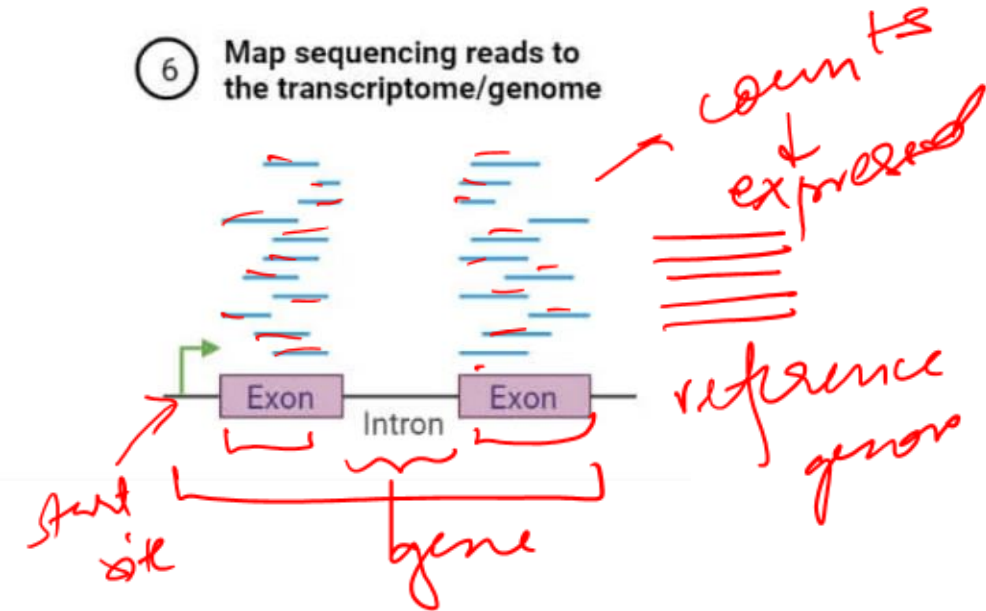
④ Ligate sequencing adapters and amplify



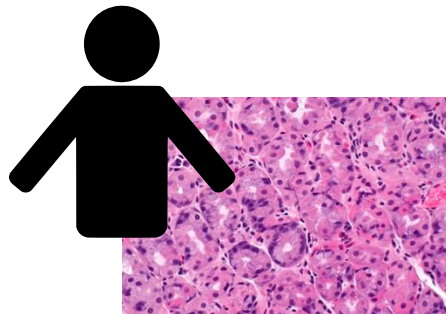
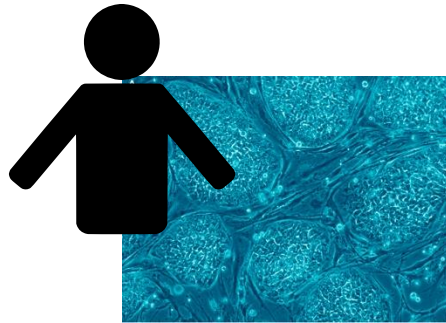
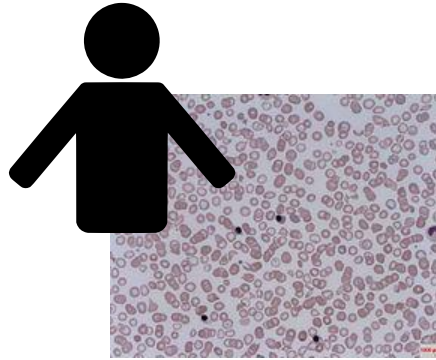
⑤ Perform NGS sequencing



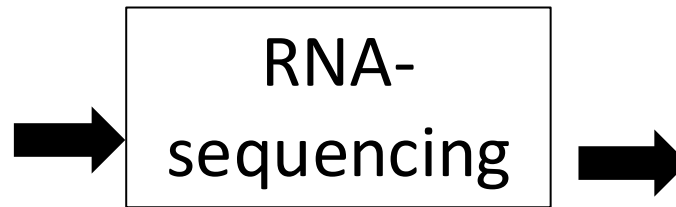
⑥ Map sequencing reads to the transcriptome/genome



Measuring gene expression for different conditions



Any questions?

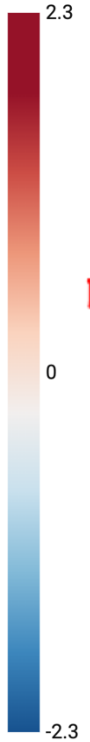
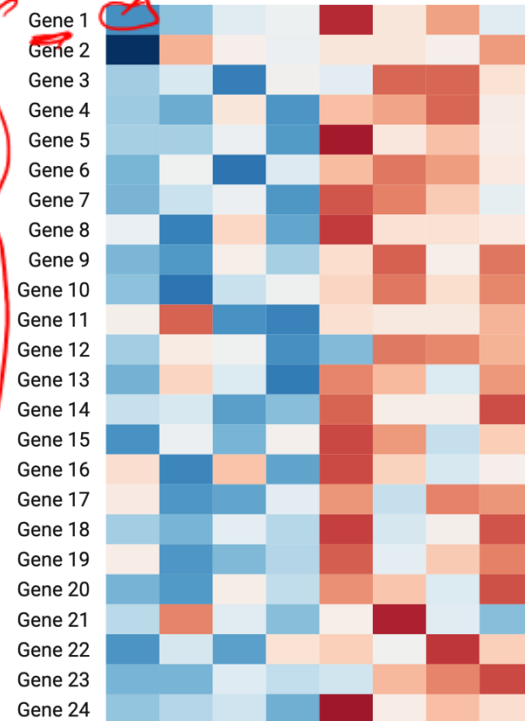


feature

Gene expression

samples

Condition 1 Condition 2 Condition 3 Condition 4 Condition 5 Condition 6 Condition 7 Condition 8



log normalized

~20,000 genes!
for humans

The Cancer Genome Atlas (TCGA)

TCGA BY THE NUMBERS

TCGA data describes



33

DIFFERENT
TUMOR TYPES

...including

10

RARE
CANCERS

...based on paired tumor and normal tissue sets
collected from



11,000

PATIENTS

...using

7

DIFFERENT
DATA TYPES

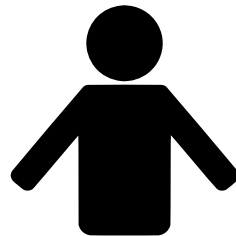


Our Problem: Cancer-type prediction

What could be the input features?

Input: $\mathbb{X}$

Gene
expression
values



What could be the output labels?

Target: $\mathbb{Y}$

What type of cancer is it?

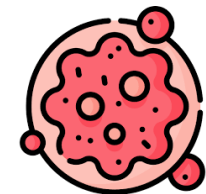


Function: f

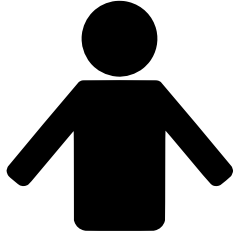
$f(\mathbb{X}) \rightarrow \mathbb{Y}$



Breast/Lung
/Prostate



Our problem: Cancer-type prediction



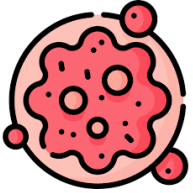
Input: $\mathbb{X} \in \mathbb{R}$

What is our input space?

What is our output space?

What is our prediction task?

Target: $\mathbb{Y}$
 $\in \mathbb{Z}$



10 samples

Gene_1	Gene_2	Gene_3	Gene_4
3.24	1.95	3.90	4.38
0.90	0.01	-0.53	-1.50
4.04	2.72	6.34	3.76
2.99	2.31	3.99	3.34
1.41	1.39	-0.88	0.08
0.88	-0.24	1.21	0.54
3.82	4.19	4.39	4.00
3.05	1.53	1.33	3.01
4.14	4.28	5.86	4.97
4.51	3.81	4.23	4.90



Function: f



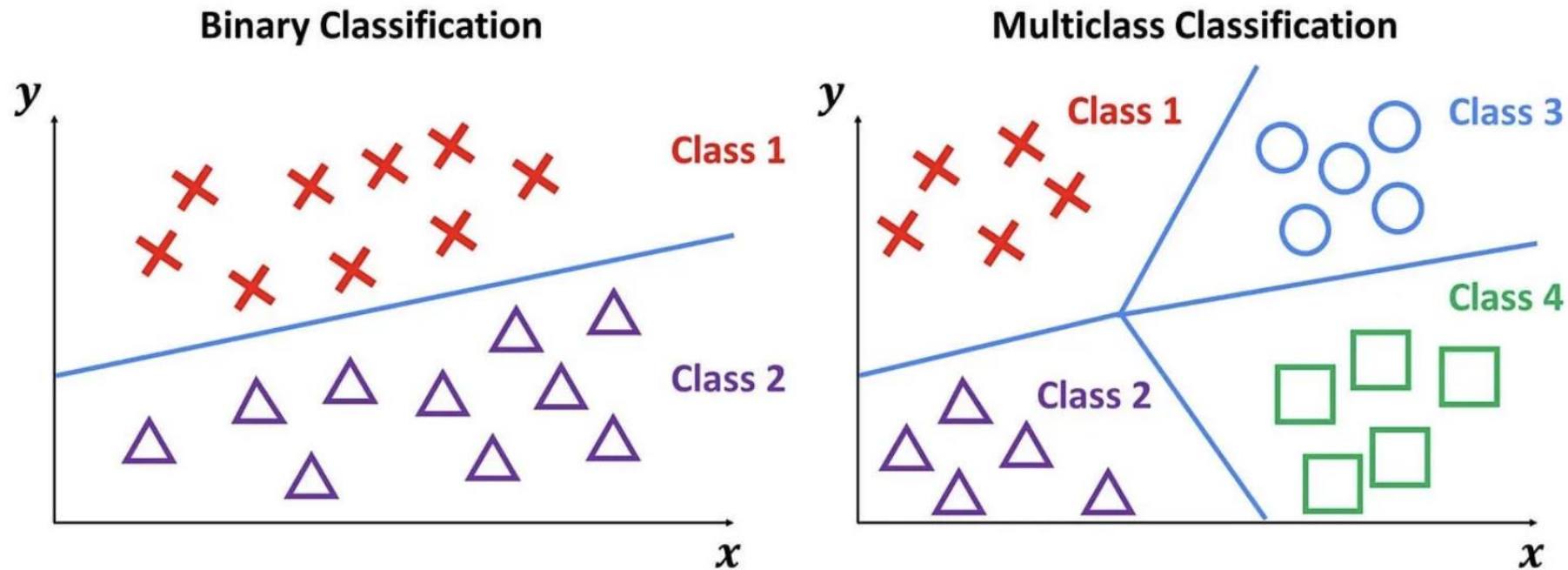
$f(\mathbb{X}) \rightarrow \mathbb{Y}$

Cancer_Type

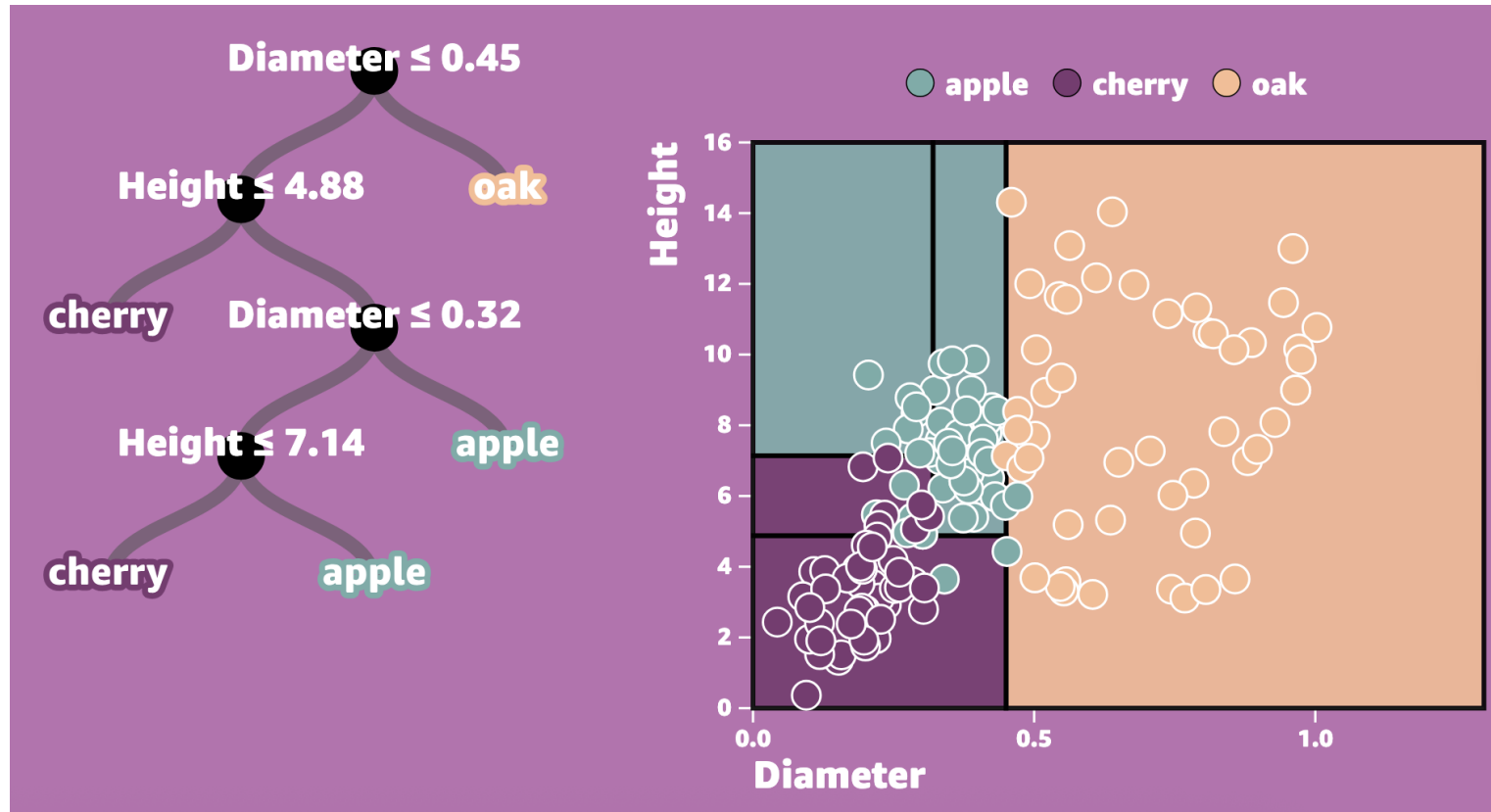
Prostate
Breast
Prostate
Prostate
Breast
Breast
Prostate
Lung
Prostate
Prostate

Multi-class classification

Multiclass classification is a machine learning task that assigns a single, exclusive label to data points from more than two distinct categories



Can random forests perform multi-class classification?

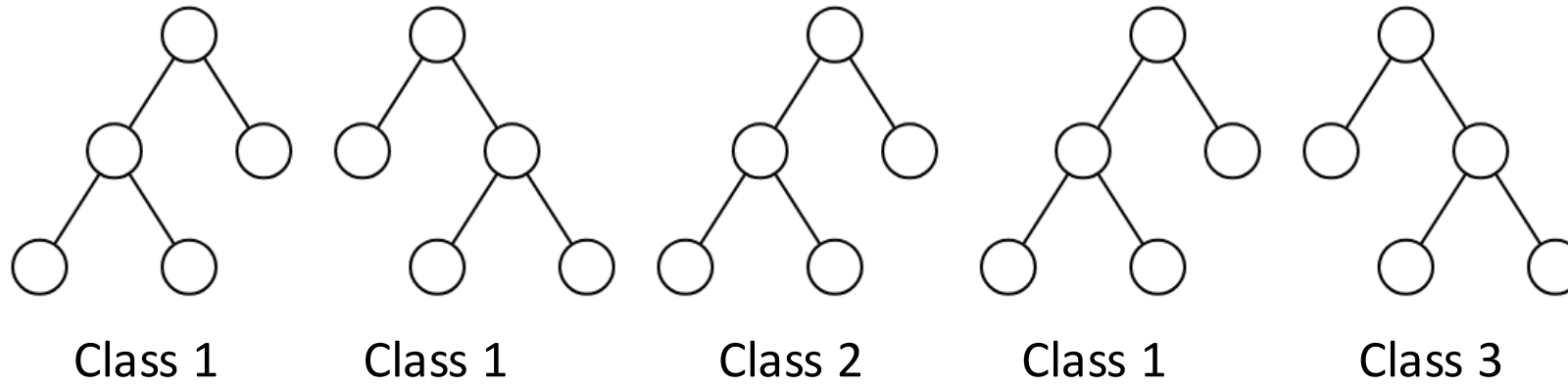


$$Gini = 1 - \sum_{k=1}^K p_k^2$$

Where:

- p_k = proportion of samples belonging to class k in that node
- K = number of classes

Can linear models do multiclass classification?



Class 1

Muddy point: How does majority voting happen mathematically?

Multi-class classification using linear models

Replace sigmoid function with ...

Softmax
function:

$$\underline{P}(y = \underline{k} \mid x) = \frac{e^{\underline{F_k}(x)}}{\sum_{j=1}^K e^{F_j(x)}}$$

$$F_k(x) = \underline{\underline{wx + b}}$$

y = true label of class k

p = predicted probability of class k

$$-(y \log p + (1-y) \log (1-p))$$

$$\mathcal{L} = - \sum_{i=1}^n \sum_{k=1}^K y_{ik} \log p_{ik}$$

type 1
type 2
type 3

1	1	0
0	0	1
0	0	0

Any questions?



Class activity

What are the classification metrics we have covered?

How do we calculate these metrics for decision trees/random forests? 

Can we use them as is for multi-class classification?

If not, then what would we need to modify?

Evaluation Metrics For Multi-Class Classification

What classification metrics do we know?

Accuracy

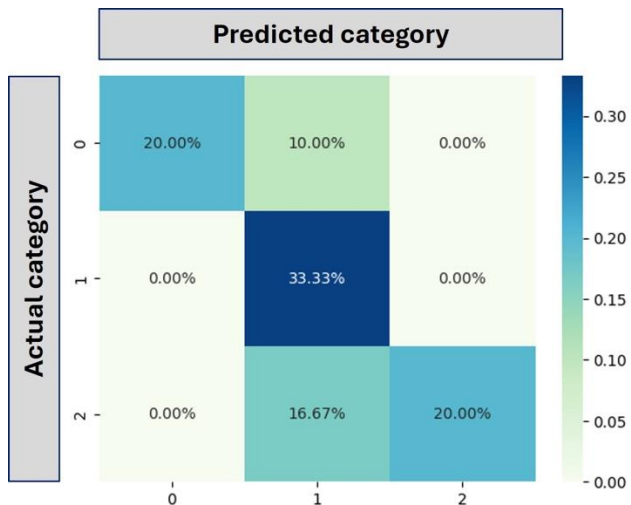
Precision = $TP / (TP + FP)$

Recall = $TP / (TP + FN)$

F1 = $2PR / (P + R)$

AUROC

AUPRC



Strategy: One versus Rest!

$$\text{Macro} = \frac{1}{K} \sum_{k=1}^K \text{Metric}_k$$

→ *weight equally*



$$\text{Weighted} = \sum_{k=1}^K w_k \cdot \text{Metric}_k$$

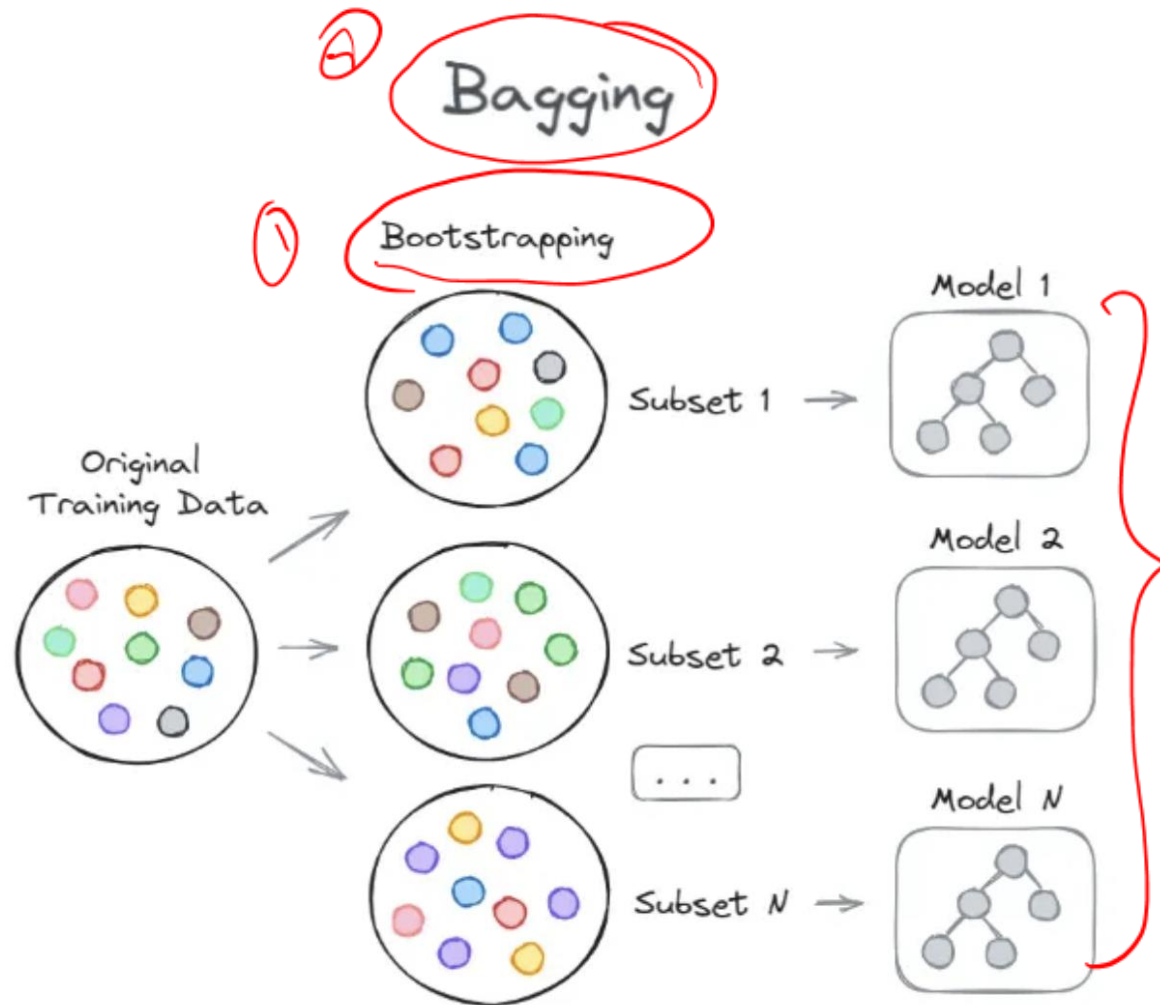
→ *favor smaller classes*

→ *favours large class*

Micro: Aggregate all predictions globally before computing metric.

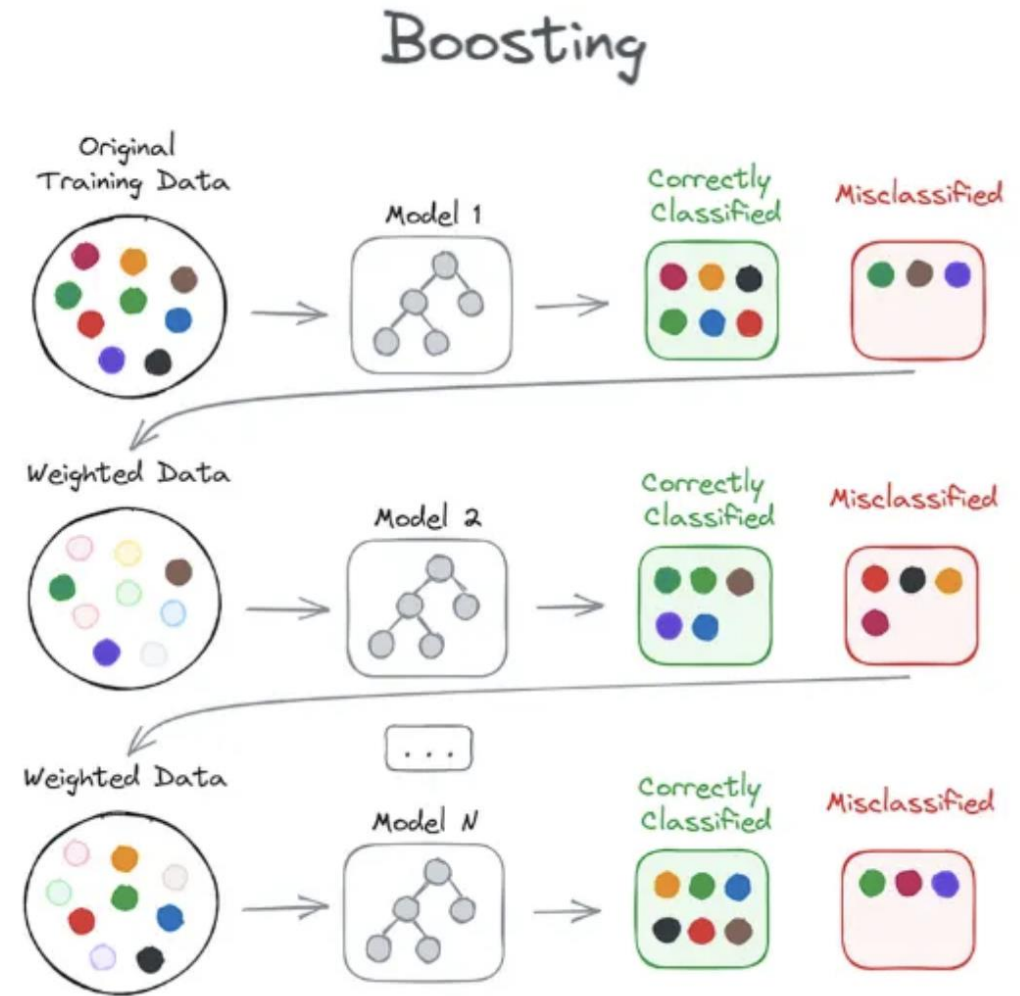
Gradient boosted trees

The three B's



Boosting

Boosting in machine learning is an ensemble technique that **sequentially builds a strong predictive model** by combining multiple "weak learners" (usually shallow decision trees), where each **new model attempts to correct the errors of its predecessor**.



Boosting

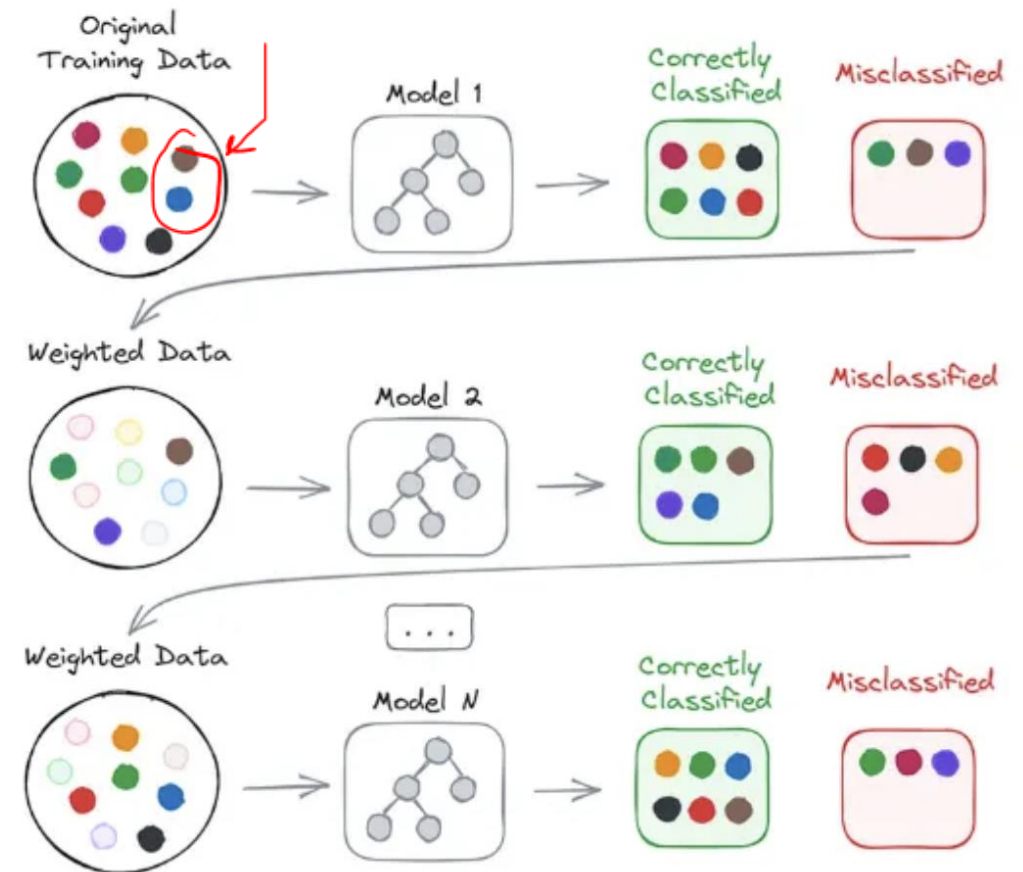
6

Adaptive boosting
→ AdaBoost
Any questions?
???



Why do boosting?

- (1) Don't have to learn strong learners, multiple weak learners instead
- (2) Easy to design computationally efficient weak learner
- (3) Can handle data imbalance



Gradient Boosting

1. Initialization:

- The ensemble starts with an initial model, often a simple one like a single-leaf tree or a constant value
- The initial prediction is made by this base model

2. Sequential Training:

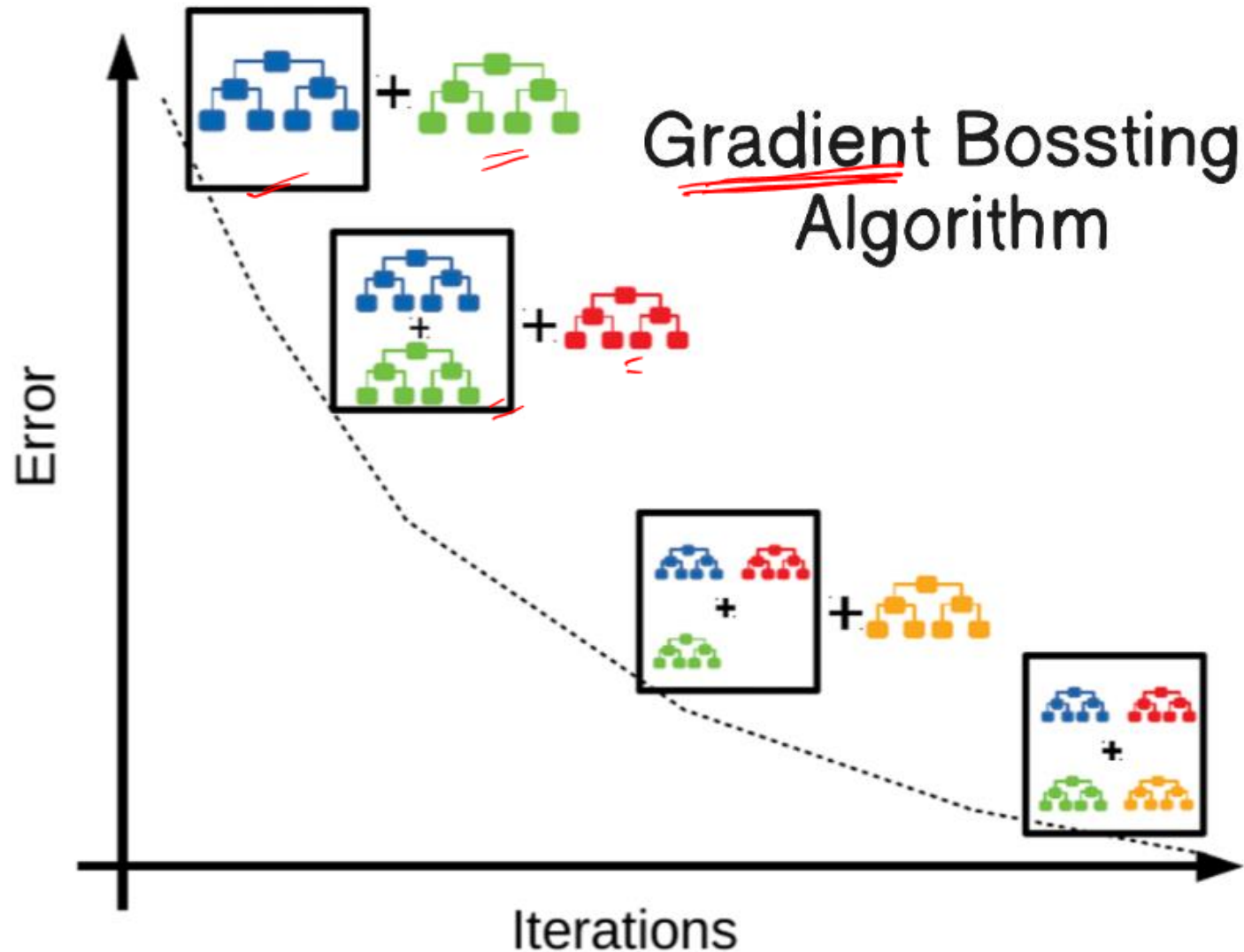
For each iteration:

- 1. Compute the residuals:** Calculate the difference between the actual target values and the predictions made by the current ensemble.
- 2. Create a new tree (Fit a base learner):** Train a new learner (typically a decision tree) on the residuals. Fit the new model to predict the residuals, rather than the original target values.
- 3. Calculate the output values:** In this part, we determine the output values for each leaf.
- 4. Make a new prediction:** Update the ensemble by adding the new model's predictions scaled by a learning rate.
- 5. Repeat** until a predefined number of iterations (trees) is reached or until a stopping criterion is met.

3. Combining Models:

- The final prediction is the sum of the initial prediction and the predictions from all subsequent models

Gradient Boosting



Note – we don't have weights and biases, so what are we updating at every iteration?

$$\rho = \frac{1}{2n} (y - F(x))^2$$
$$\frac{\partial \rho}{\partial F(x)} = y - F(x)$$

i	x	y	$F_0(x)$	res	$F_1(x)$	res	$F_2(x)$
1	1	2	5	-3	3	-1	2
2	2	4	5	-1	3	+1	4
3	3	6	5	+1	7	-1	6
4	4	8	5	+3	7	+1	8

$$y = 2x$$

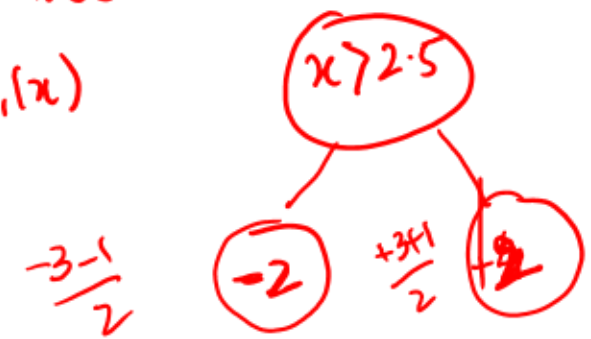
① Initialize a simple model

$$F_0(x) = \bar{y} = \frac{2+4+6+8}{4} = 5$$

② $r_i = y_i - F_0(x)$ } gradient of squared error
calculate residual

③ Fit tree to residuals

$h_1(x)$

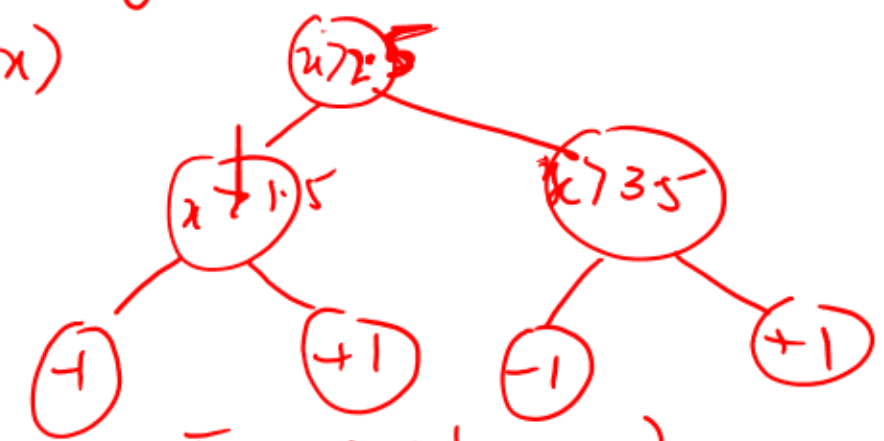


④ Output, update model

$$F_1(x) = F_0(x) + h_1(x)$$

⑤ $r_i = y_i - F_1(x)$

⑥ $h_2(x)$



⑦
$$F_2(x) = F_1(x) + h_2(x)$$

$$= \underbrace{F_0(x)} + \underbrace{h_1(x)} + \underbrace{h_2(x)}$$

Recap - Learn about multi-class classification and boosting

(1) Introducing the task – Predicting cancer-type using gene expression

Homework Reading:
[Chapter 13 in CIML Book](#)

(2) Multi-class classification

(3) Class Activity: Multi-class classification evaluation

Kaggle article on evaluation metrics:
<https://www.kaggle.com/code/nkitgupta/evaluation-metrics-for-multi-class-classification>

(4) Boosting

(5) Gradient Boosted Trees

Wrap up



What was the clearest point today?

What was the muddiest point today?

